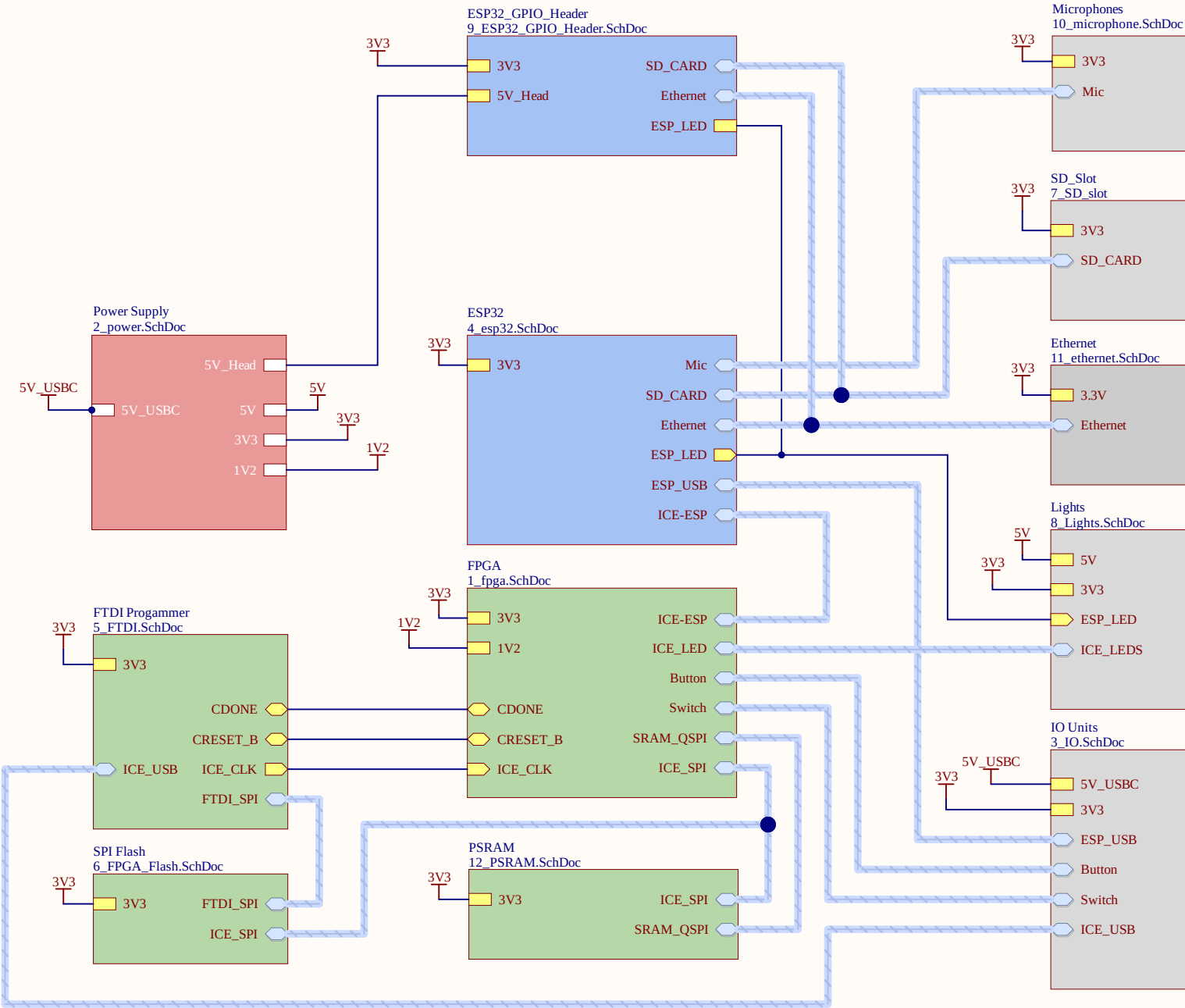
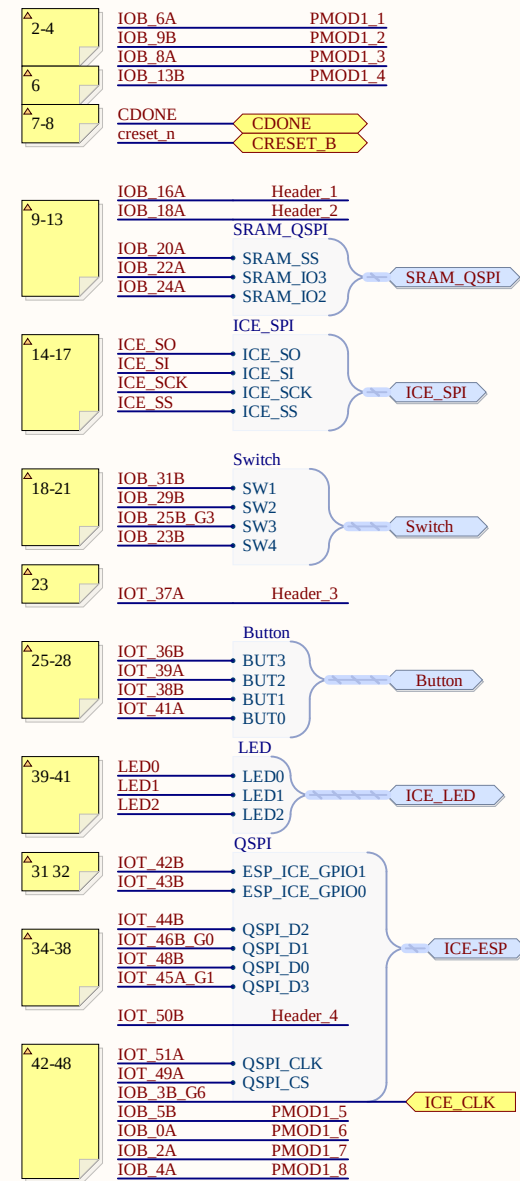
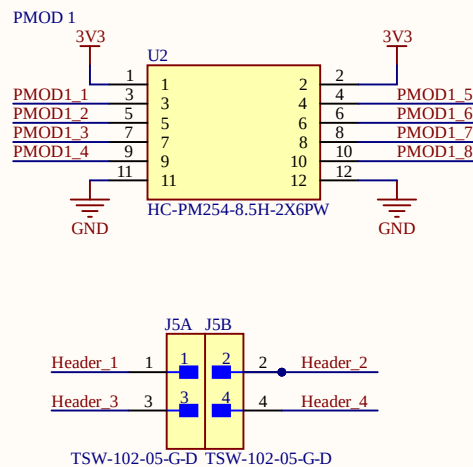
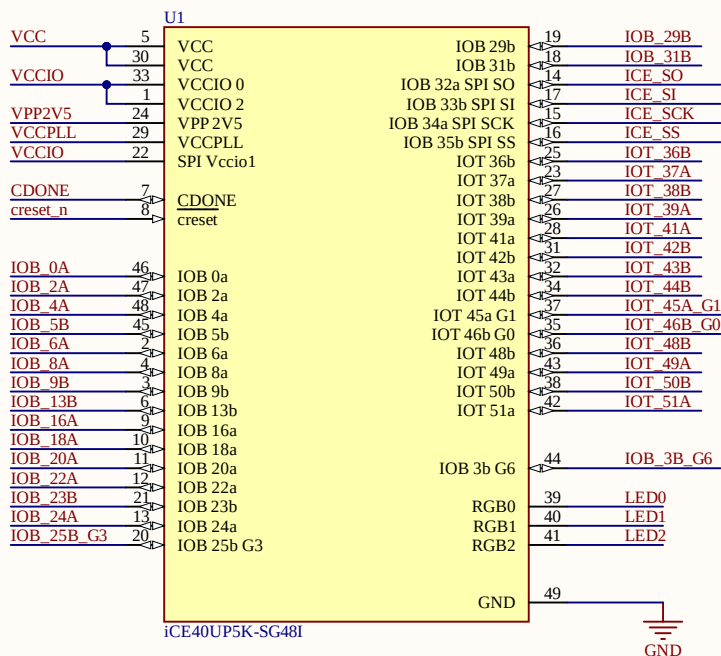
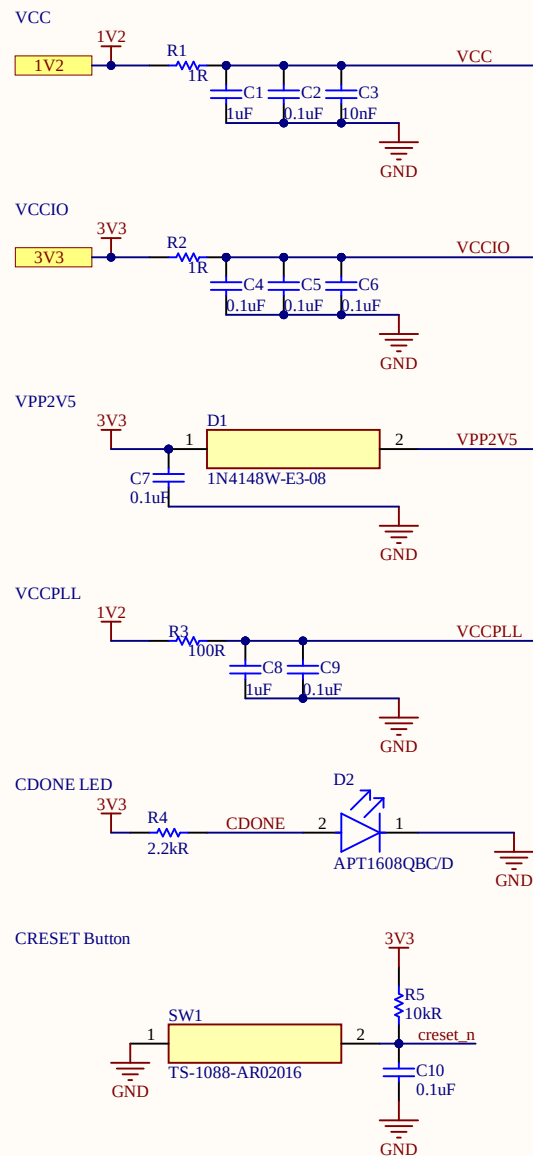




KC-1 FPGA MCU Devboard

A3	Project:	Revision: *
Date: 2026-09-26	Time: 10:54:22 PM	Sheet 1 of 13
File: 1_reference.SchDoc	Drawn By: Albert Huang, Matthew Kong, A	



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A4

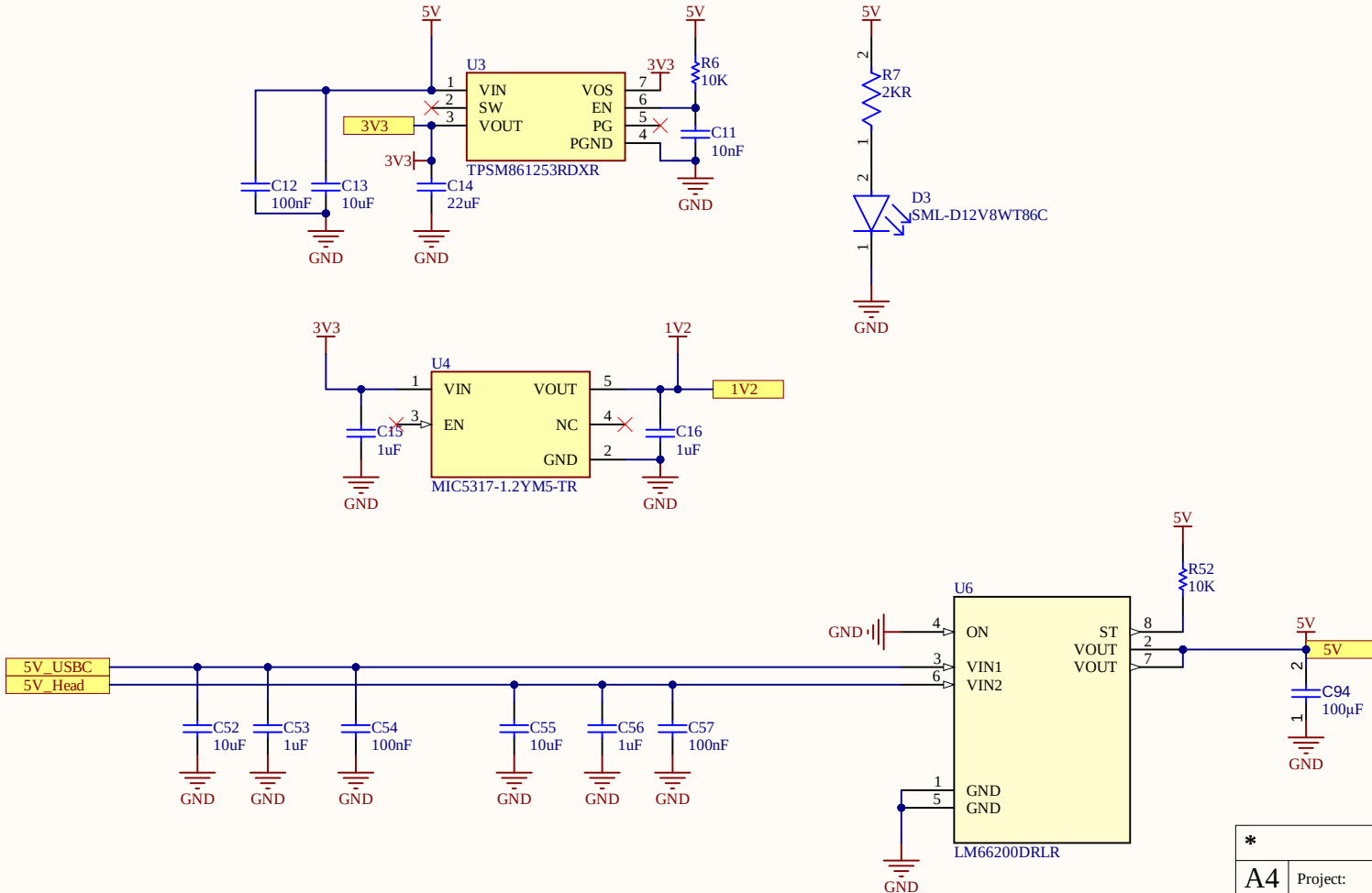
Project:

Revision: *

Date: 2026-09-26 Time: 10:54:22 PM Sheet 2 of 13

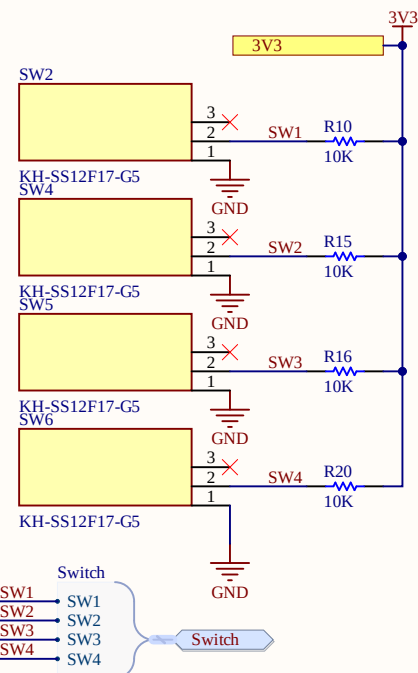
File: 1_fpga.SchDoc

Drawn By: *

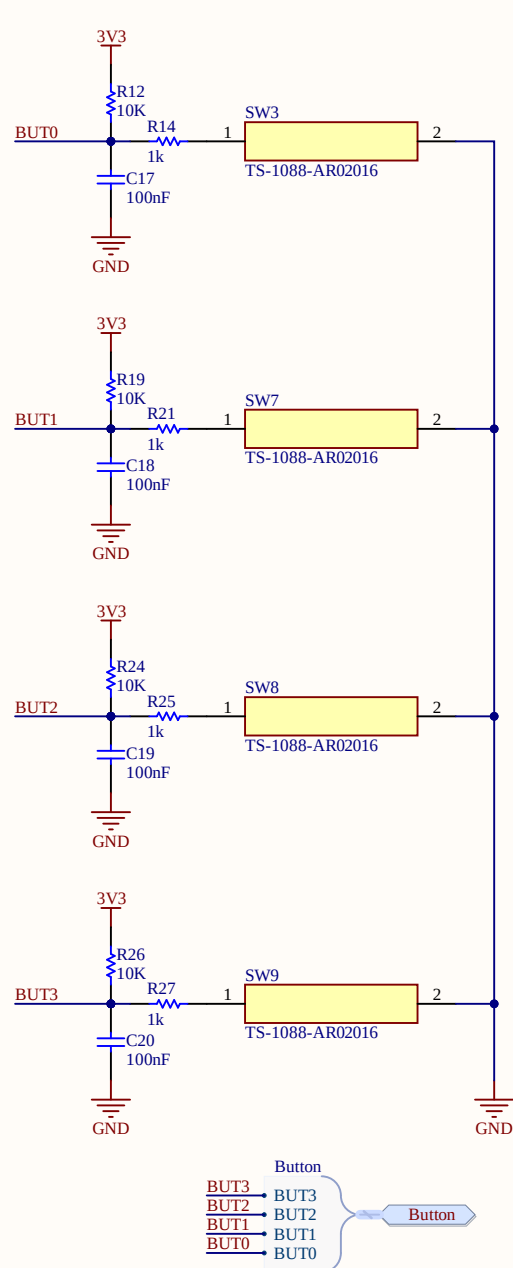


*			
A4	Project:		Revision: *
Date:	2026-09-26	Time: 10:54:22 PM	Sheet 3 of 13
File:	2_power.SchDoc		Drawn By: *

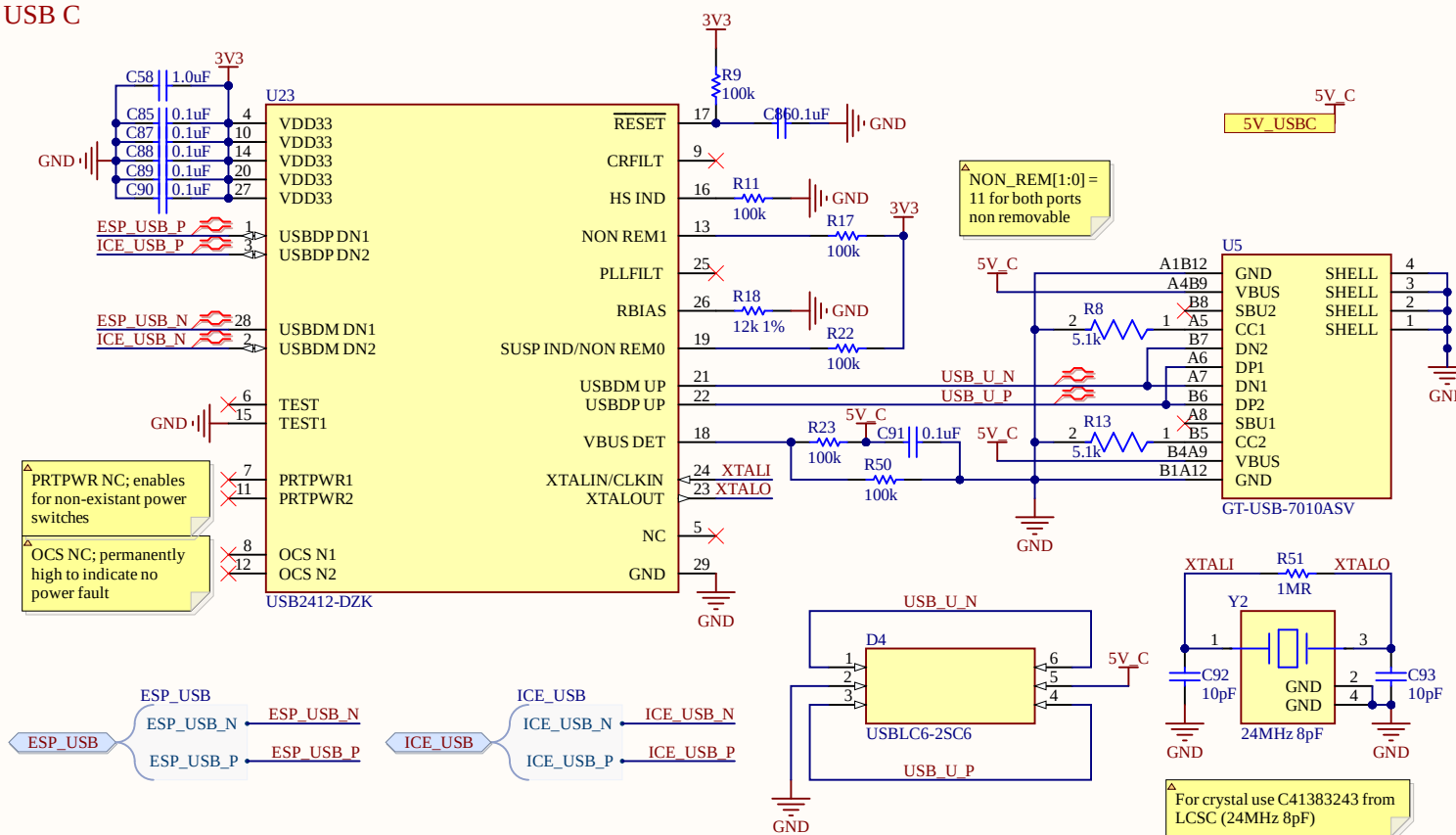
Switches

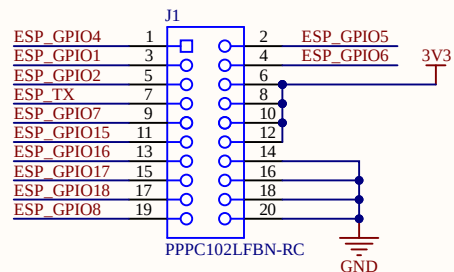


Push Buttons



USB C





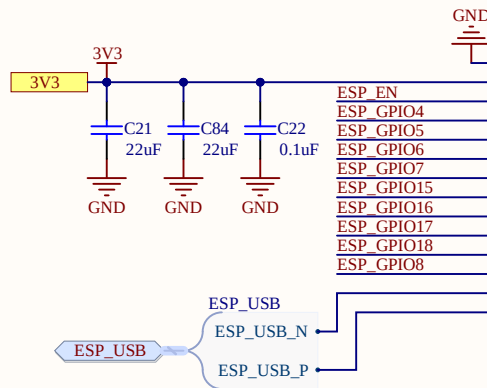
△ All of these can be used as normal GPIO

△ I2C - any pins, only 2 I2C buses at once tho

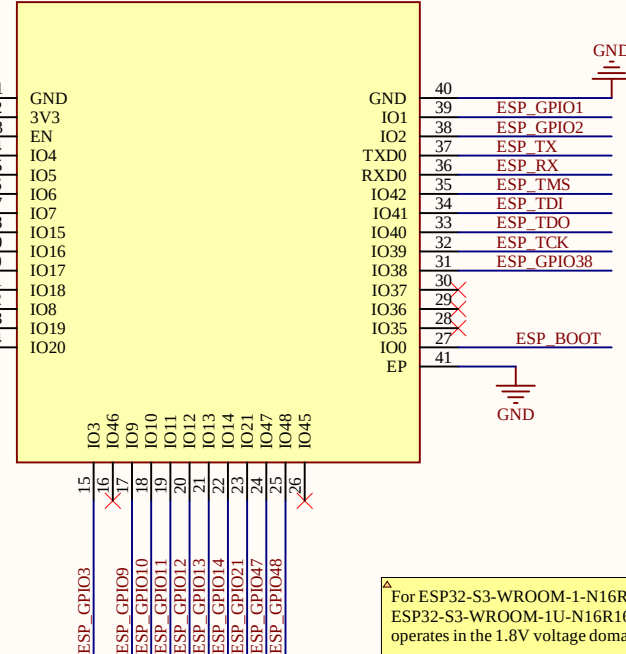
△ SPI2 - IO9-14
SPI3 - any

△ UART - use TXD0 and RXD0 pins, otherwise any

△ available: 4, 5, 6



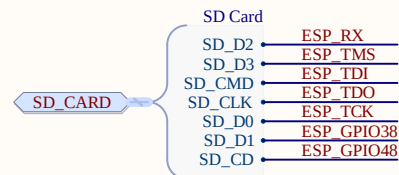
U7 ESP32-S3-WROOM-1-N8



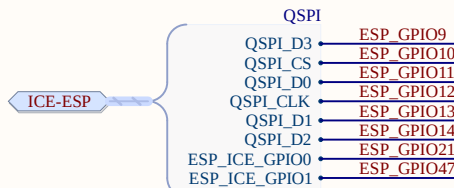
△ For ESP32-S3-WROOM-1-N16R16V and ESP32-S3-WROOM-1U-N16R16V, IO47/IO48 operates in the 1.8V voltage domain



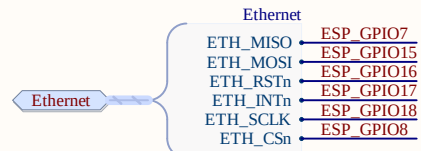
△ I2S



△ GPIO

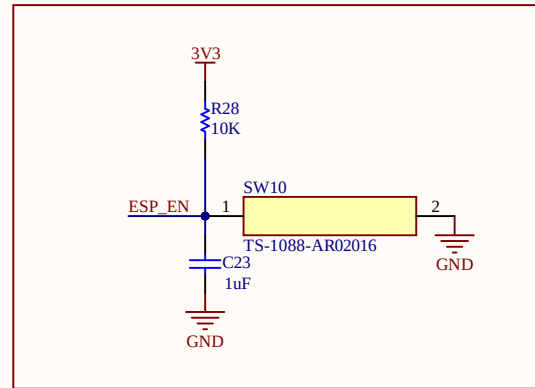


△ SPI2

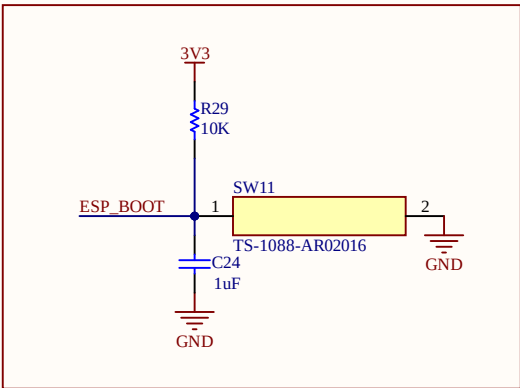


△ SPI3

EN/Reset



BOOT



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A4

Project:

Revision: *

Date: 2026-09-26

Time: 10:54:22 PM

Sheet 5 of 13

File: 4_esp32.SchDoc

Drawn By: *

A

A

B

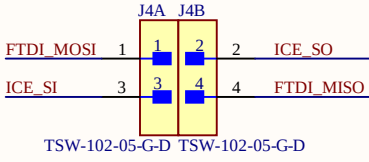
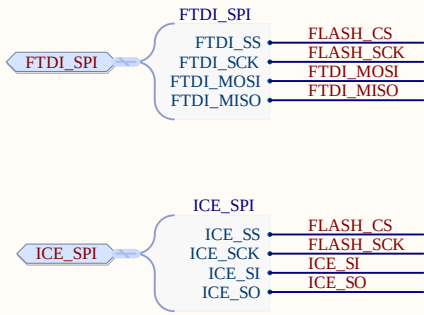
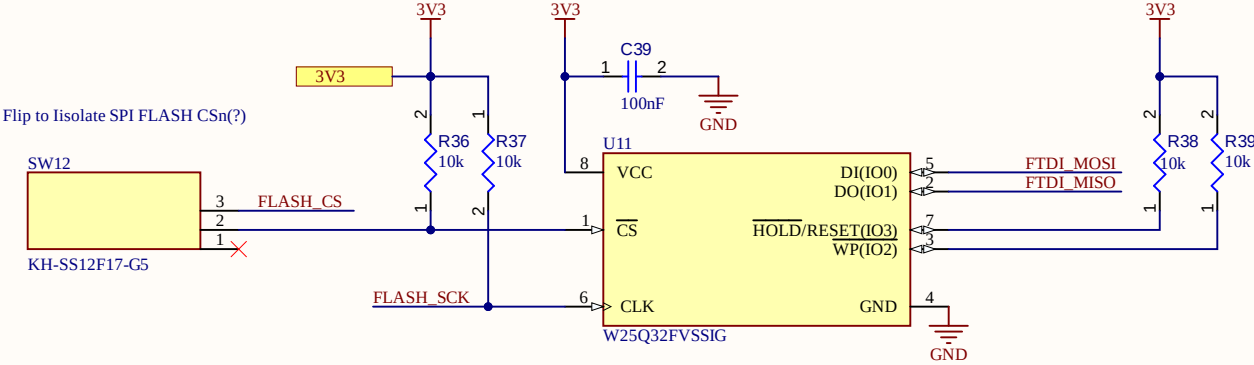
B

C

C

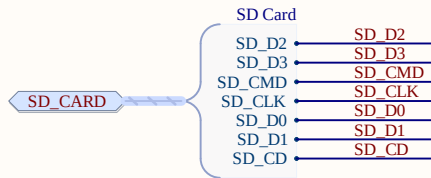
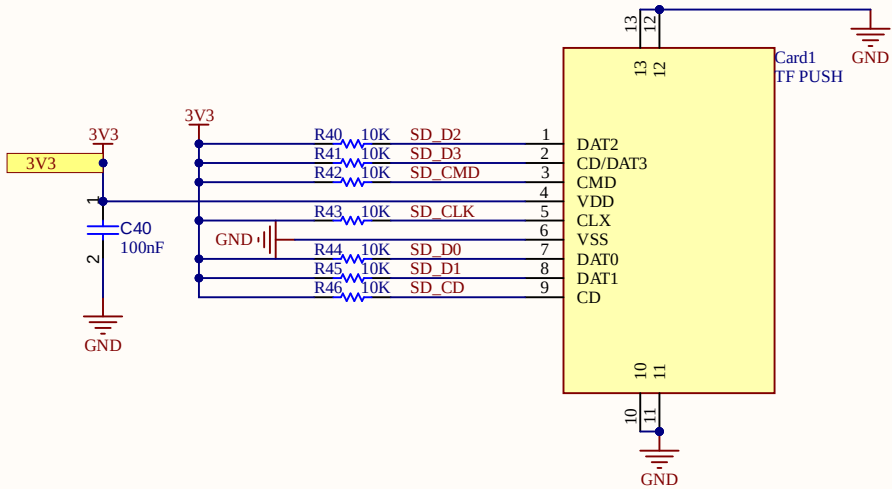
D

D

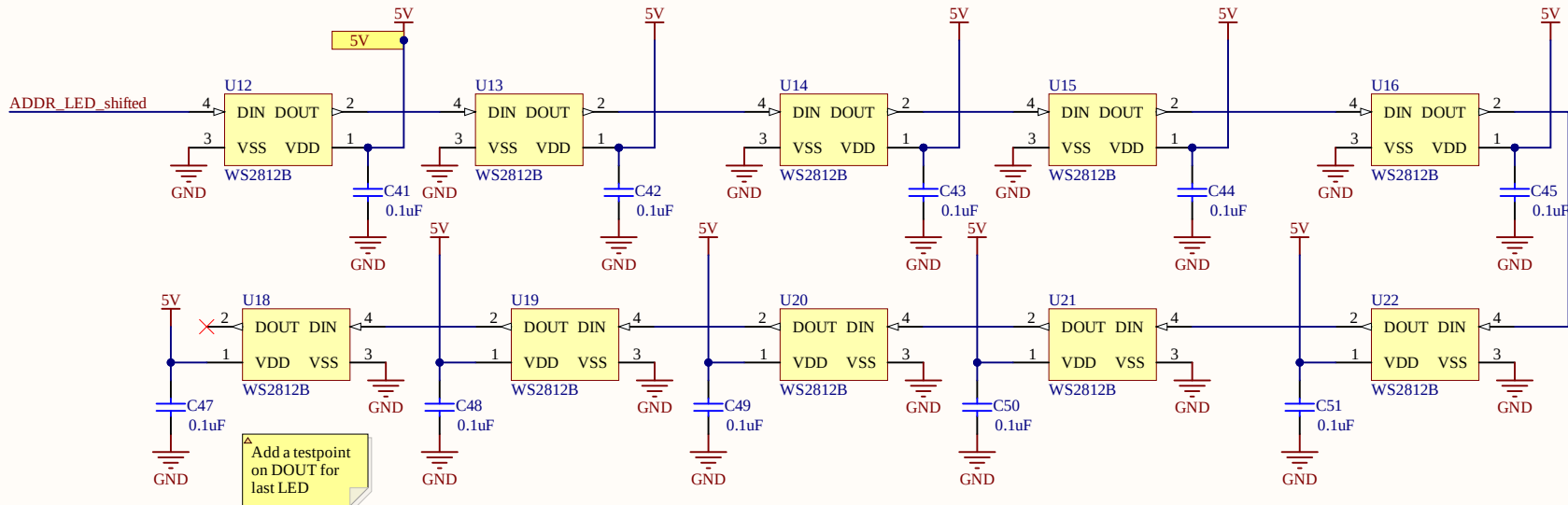


Δ = For programming Flash -
Shunt 1,2 and 3,4 (default)
FTDI: Master, Flash:
Slave.
CRESET_B = low

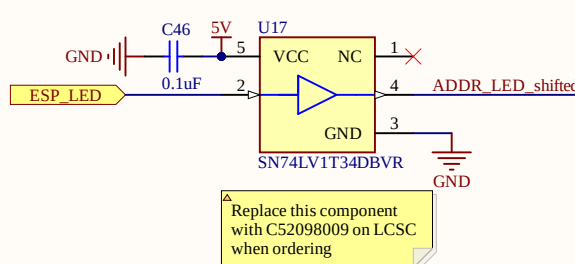
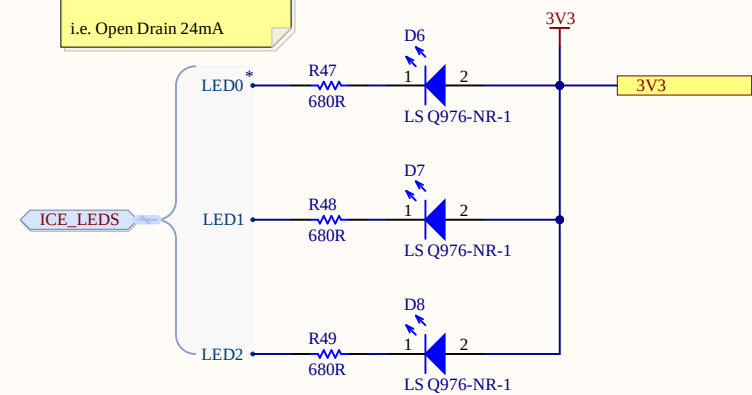
|| For programming iCE -
Shunt 1,3 and 2,4
FTDI: Master, ICE: Slave



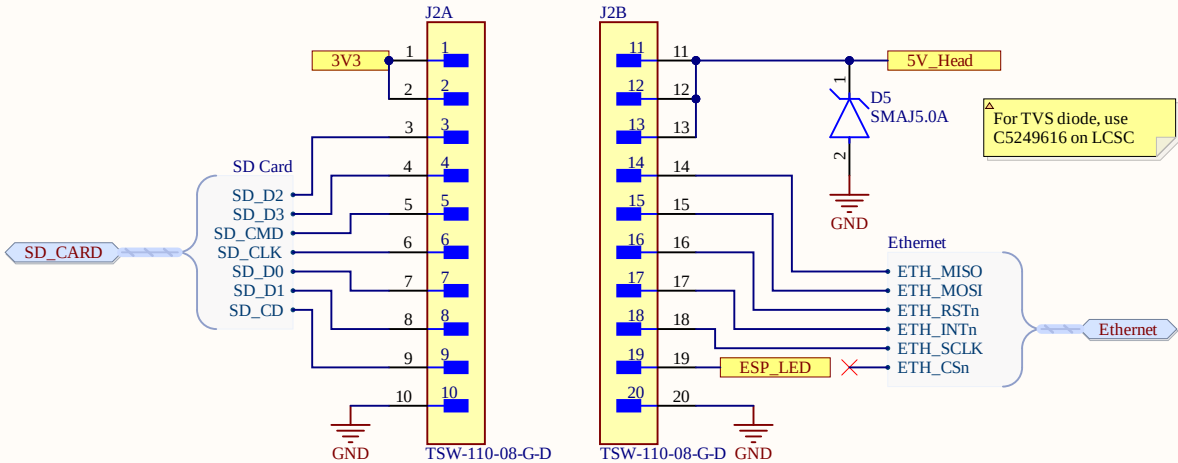
*			
A4	Project:		Revision: *
Date: 2026-09-26	Time: 10:54:22 PM	Sheet 8 of 13	
File: 7_SD_slot.SchDoc		Drawn By: *	



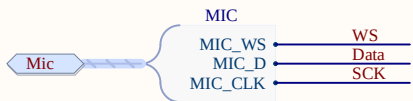
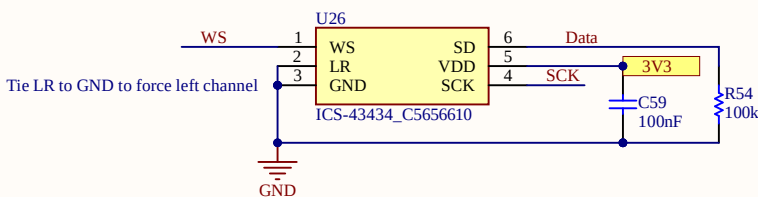
Double Check with the ICE pin electric characteristic.
i.e. Open Drain 24mA



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A4	Project:	Revision: *	
Date:	2026-09-26	Time: 10:54:22 PM	Sheet 9 of 13
File:	8_Lights.SchDoc	Drawn By: *	



*			
A4	Project:		Revision: *
Date:	2026-09-26	Time: 10:54:22 PM	Sheet 10 of 13
File:	9_ESP32_GPIO_header.SchDoc		Drawn By: *



- 1 WS Input Serial Data-Word Select for I²S Interface. This pin includes an internal 100 k Ω pull-down resistor. If the pin is pulled to VDD in standby mode, then VDD/100 k Ω current will sink through the pin.
- 2 LR Input Left/Right channel select. When set low, the microphone outputs its signal in the left channel of the I²S frame. When set high, the microphone outputs its signal in the right channel.
- 3 GND Ground Ground. Connect to ground on the PCB.
- 4 SCK Input Serial Data Clock for I²S Interface.
- 5 VDD Power Power, 1.65 to 3.63 V. This pin should be decoupled to GND with a 0.1 μ F capacitor.
- 6 SD Output Serial Data Output for I²S Interface. This pin tristates when not actively driving the appropriate output channel. The SD trace should have a 100 k Ω pull-down resistor to discharge the line during the time that all microphones on the bus have tristated their outputs

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A4	Project:		Revision: *
Date:	2026-09-26	Time: 10:54:22 PM	Sheet 11 of 13
File:	10_microphone.SchDoc		Drawn By: *

